

Multi-Objective Asset-Based Weapon-Target Assignment Modeling For Cost-Effective Defense Against Loitering Munition Swarms

Maksymilian Gach

*Department of Advanced Computing Sciences
Faculty of Science and Engineering
Maastricht University
Maastricht, The Netherlands*

Abstract—This thesis addresses the economic and computational challenges posed by loitering munition swarms in modern air defense. Extreme cost asymmetry between inexpensive drones and high-value interceptors renders the traditional kinetic-maximization strategies unsustainable, necessitating more cost-aware resource allocation logic. To address this issue, a multi-objective Asset-Based Weapon-Target Assignment (AB-WTA) framework was developed. The framework evaluates the Non-dominated Sorting Genetic Algorithm II (NSGA-II) against a Greedy Maximum Marginal Return baseline and a random search, testing computational scalability against simulated swarms of up to 1,000 threats. Empirical results demonstrate that the NSGA-II algorithm achieved the execution latency required for real-time tactical response, successfully generating Pareto-optimal strategies that conserve interceptors while minimizing infrastructure loss. Furthermore, evaluating these assignments through an adversarial inference model revealed that the NSGA-II metaheuristic significantly mitigated information leakage regarding defended asset valuations.

Index Terms—Weapon-Target Assignment, Air Defense, Loitering Munitions, Multi-Objective Optimization, NSGA-II, Resource Allocation.

I. INTRODUCTION

The field of modern aerial warfare is currently undergoing a fundamental transformation, driven by the proliferation of autonomous loitering munitions. Historically, air defense was designed to counter high-value targets, such as manned aircraft or cruise missiles, using sophisticated surface-to-air missiles (SAM) in order to protect valuable ground assets. This approach heavily relied on the assumption that threats would be relatively few and of high individual cost, making the interception economically viable.

The emergence of unmanned aerial vehicles (UAVs) has inevitably shifted this logic. These “kamikaze drones” are inexpensive to produce and thus can be deployed in massive quantities. In recent conflicts, these platforms have been used ever more frequently to overwhelm traditional Integrated Air Defense Systems (IADS) through saturation attacks [1].

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The primary challenge is no longer just the kinetic ability to intercept a target, but the economic sustainability of such operations. When an adversary utilizes a swarm of drones costing a few thousand dollars each to deplete a defender’s stock of interceptors costing millions, the defender faces a strategic long-term challenge [2].

To address the challenge of allocating defensive resources, military planners rely on the WTA modeling of the battlefield. Fundamentally, the WTA problem seeks the optimal assignment of a limited inventory of weapons to a set of incoming threats to maximize kill probability. Mathematically, it is formulated as an NP-hard combinatorial optimization problem [3], necessitating the deployment of highly efficient, scalable algorithms to generate actionable defensive assignments within strict tactical timeframes.

Traditionally, defense systems have utilized Target-Based WTA models, which focus strictly on maximizing the expected destruction of incoming threats [4]. However, against saturation attacks where complete interception is logistically and physically impossible, this approach is severely flawed. Instead, modern defense requires a multi-objective Asset-Based WTA (MO-ABWTA) approach [5]. Rather than maximizing drone kills, an MO-ABWTA framework can balance two critical priorities: the survival of high-value ground assets and the active conservation of expensive interceptor stockpiles. It recognizes that in a massive swarm attack, some threats must be allowed to pass to ensure that limited defensive resources are strategically reserved for threats directly endangering the most critical assets. This specific multi-objective formulation will be referred to simply as the AB-WTA framework throughout the remainder of this thesis, directly addressing the primary economic vulnerability of traditional defense.

The secondary vulnerability of current approaches is strictly informational. Because these single-objective heuristics (such as greedy algorithms) prioritize threats based on strict, static rules, they are highly predictable. If a defense system consistently executes an optimal, deterministic strategy, an observing adversary can utilize the concept of “Revealed Preferences” [6] to reverse-engineer the defender’s hidden priorities. By observing where a defender expends expensive interceptors,

the adversary can deduce the true valuations of ground assets that are protected, refining subsequent waves to target newly identified vulnerabilities.

To bridge these gaps, this thesis proposes the utilization of a multi-objective Genetic Algorithm—specifically the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [7] to solve the formulated AB-WTA problem. By introducing engagement cost as an additional objective alongside asset survival, the framework generates an optimal Pareto front of weapon-target assignments. This allows decision makers to navigate the primary economic trade-off between kinetic protection and strategic ammunition conservation. Furthermore, this thesis investigates the secondary tactical utility of the Pareto front: by actively alternating optimal assignments, the defender can introduce operational obfuscation, masking true asset values from adversarial inference.

To systematically investigate this approach, this thesis addresses the following research questions:

- **RQ1:** How efficiently does the NSGA-II algorithm navigate the highly constrained AB-WTA search space compared to a random search and greedy baseline?
- **RQ2:** How do Pareto-optimal assignment strategies balance the financial trade-off between interceptor depletion and expected asset loss during asymmetric saturation drone attacks?
- **RQ3:** To what extent does utilizing multi-objective assignment strategies obfuscate true asset valuations from adversarial inference compared to predictable, single-objective greedy assignments?

To isolate the core economic and combinatorial logic, this framework models the Static Weapon-Target Assignment (SWTA) problem [8], evaluating a discrete battlefield snapshot rather than continuous dynamics. Adversarial production costs and complex physical variables are explicitly excluded. Crucially, while the developed software architecture and the model formulation fully support geographic coordinates and engagement ranges, these spatial constraints are disabled during experiments. The optimization is driven strictly by interceptor capacities, engagement costs and economic trade-offs, leaving probabilistic threat mapping for future extensions.

The remainder of this thesis is structured as follows. Section II details the mathematical formulation and methodology of the AB-WTA framework. Section III outlines the experimental design and procedural generation. Section IV presents the empirical results, which are interpreted in Section V. Finally, Section VI concludes the research and discusses broader implications.

II. METHODS

A. Mathematical Formulation

To represent the aerial battlefield scenario, the framework defines the following notation, with the subsequent objective functions and system constraints conceptually inspired by the methodologies aggregated in [8]:

- W : the set of weapon batteries
- T : the set of incoming targets (drones)
- A : the set of protected ground assets
- V_k : the value of asset k
- C_i : the cost of an interceptor from battery i
- W_i : the interceptor capacity of battery i
- p_{ij} : the probability weapon i destroys threat j
- T_k : the set of threats targeting asset k
- x_{ij} : the number of interceptors assigned from i to j

B. Objective Functions

Consequently, the framework evaluates each candidate decision matrix X against two objective functions:

Objective 1: Minimizing Engagement Cost

The first objective quantifies the cost of all interceptors used for the defense. It is formulated as a linear function subject to minimization:

$$\min F_1(X) = \sum_{i=1}^{|W|} \sum_{j=1}^{|T|} C_i x_{ij} \quad (1)$$

Objective 2: Minimizing Expected Asset Loss

The second objective models the potential asset loss caused by threats successfully striking their target. An asset k is assumed to survive if all threats targeting it ($j \in T_k$) are successfully intercepted. Assuming independent engagements, the probability that a single threat j survives x_{ij} intercepts from battery i is $(1 - p_{ij})^{x_{ij}}$ [8]. Therefore, the expected economical loss is defined as:

$$\min F_2(X) = \sum_{k=1}^{|A|} V_k \left(1 - \prod_{j \in T_k} \left(1 - \prod_{i=1}^{|W|} (1 - p_{ij})^{x_{ij}} \right) \right) \quad (2)$$

The important premise of this formulation is that the destruction of a ground asset is treated as binary and absolute. If even a single threat successfully breaches the defense and impacts its target, 100% of the asset's value (V_k) is considered lost. Consequently, the objective function F_2 naturally caps the maximum possible loss per asset at V_k .

C. System Constraints

The optimization of $F_1(X)$ and $F_2(X)$ is strictly bound by the physical limitations of the defending weapon systems. The decision matrix X must satisfy the following constraints:

Constraint 1: Interceptor Capacity

The total number of interceptors assigned from a specific battery i cannot exceed its ammunition capacity W_i .

$$\sum_{j=1}^{|T|} x_{ij} \leq W_i \quad \forall i \in \{1, \dots, |W|\} \quad (3)$$

Constraint 2: Domain

Interceptors are discrete physical entities; therefore, the decision variable must be a non-negative integer.

$$x_{ij} \in \mathbb{Z}_{\geq 0} \quad \forall i \in \{1, \dots, |W|\}, \forall j \in \{1, \dots, |T|\} \quad (4)$$

D. Baseline Algorithms

To establish analytical benchmarks for both computational efficiency (RQ1) and strategic ambiguity (RQ3), the NSGA-II framework is evaluated against two baseline solvers:

Greedy Maximum Marginal Return (MMR):

To represent traditional, deterministic air defense doctrine, this framework implements a Greedy MMR heuristic [9]. By iterating through a discrete range of preference weights $w \in [0, 1]$, this method is adapted to generate a baseline Pareto front. Operating on a strict logic of weighted maximization, the algorithm evaluates all valid interceptor-threat pairings to identify the assignment yielding the greatest absolute improvement: $w \cdot \Delta F_2 - (1 - w) \cdot C_i$, where ΔF_2 is the marginal reduction in expected asset loss and C_i is the interceptor cost. This loop repeats until ammunition is exhausted, threats are neutralized, or the marginal improvement falls below zero, signifying that any further interception is no longer cost-effective.

Uniform Random Search:

To demonstrate that the genetic algorithm is actively learning the fitness landscape rather than benefiting from a purely stochastic distribution, a random-search baseline is implemented. This "solver" generates randomized, perfectly legal decision matrices using the exact same bounds as the NSGA-II chromosome. To maintain equivalent computational effort, Random Search evaluates the same number of candidate solutions as the most extensive NSGA-II configuration (20,000 evaluations).

E. The NSGA-II Optimization Framework

The proposed framework uses the Non-dominated Sorting Genetic Algorithm II (NSGA-II) as the primary solution method for the defined problem. NSGA-II is a population-based metaheuristic that is widely recognized as a benchmark in WTA literature due to its demonstrated capacity to maintain a diverse and well-distributed Pareto front [10].

A computational challenge in applying genetic algorithms to the AB-WTA problem is ensuring that the evolutionary process, such as crossover and mutation, does not generate any illegal solutions that violate any of the described constraints. Traditional penalty functions waste valuable evaluation resources on dead-end assignments. To eliminate this inefficiency, the framework employs a constraint-free, one-dimensional integer array encoding rather than evolving the two-dimensional decision matrix X directly [11].

The length of the chromosome is equal to the total number of available interceptors across all the batteries ($\sum_{i=1}^{|W|} W_i$). Each gene represents a specific physical interceptor, and its value defines the assigned target ID j . To physically implement the framework's ammunition conservation logic, the target domain is expanded to include a dummy value, $|T|$, acting

as a "hold fire" encoding. To accurately calculate engagement costs, a static lookup array maps each gene index back to its specific originating battery. This approach supports weapon heterogeneity.

A helper function translates this one-dimensional chromosome back into the two-dimensional decision matrix X for fitness evaluation. It also serves as a feasibility mask, actively filtering out invalid assignments. Its inclusion, combined with the capacity-bound chromosome length, guarantees a strictly valid evolutionary population.

F. Evaluation Metric for Strategic Ambiguity

To answer RQ3 and investigate the secondary tactical utility of the multi-objective framework, this methodology introduces the "Informed Observer" model. It's meant to simulate adversarial intelligence gathering processes based on the economic concept of Revealed Preferences [12], aiming to quantify the extent of information leakage during assigned engagements.

Against traditional Target-Based defenses, an adversary might infer asset value simply by counting the number of interceptors fired towards each munition. However, against a cost-aware, Asset-Based defense, this naive logic fails. In saturation scenarios, a defender might intentionally abandon a high-value asset targeted by an overwhelming number of threats if the cost of interception exceeds the expected asset loss. Conversely, the defender might readily intercept a single threat targeting a lower-value asset if the engagement yields a positive marginal return.

Therefore, the adversary is modeled as an informed, rational observer who knows the financial expenditure of each interception (C_i) and its kill probability (p_{ij}). The observer assumes that a rational defender allocates resources such that the financial investment deployed to protect an asset is proportional to its hidden value (V_k). The adversary's estimated value E_k for a given asset is calculated as the total financial cost of all allocated interceptors:

$$E_k = \sum_{j \in T_k} \sum_{i=1}^{|W|} C_i x_{ij} \quad (5)$$

To quantify the degree of information leakage, this framework utilizes Spearman's Rank Correlation Coefficient (r_s) [13]. The true asset values (V_k) and the adversary's estimates (E_k) are converted into comparative rank arrays. The statistical correlation between these two ranked arrays is calculated as:

$$r_s = 1 - \frac{6 \sum_{k=1}^{|A|} d_k^2}{|A|(|A|^2 - 1)} \quad (6)$$

The resulting coefficient r_s falls within the continuous range of $[-1.0, 1.0]$. A coefficient approaching 1.0 would indicate a strong positive correlation, meaning the allocation scales accurately its financial response to the asset's true value, allowing the adversary to successfully reverse-engineer the defender's exact priority list. A coefficient approaching 0.0 indicates a complete lack of statistical correlation, blinding the adversary's inference model.

G. Software Framework and Implementation

To operationalize the mathematical models and optimization algorithms described above, a custom, object-oriented software framework was developed entirely in Python, with the complete source code made publicly available [14]. This framework encapsulates the battlefield parameters and objective functions into a simulation environment. It allows the systematic comparison of solution assignments across given or generated scenarios.

A key architectural feature is the strict isolation of the core AB-WTA simulation logic from the optimization solvers. This modularity ensures the straightforward integration of new solvers and objective benchmarking between existing ones.

The genetic algorithm operations were implemented utilizing the `pymoo` optimization library [15], while core matrix operations and joint probability calculations were heavily vectorized using `NumPy` [16].

III. EXPERIMENTS

The experimental section of this research is designed to empirically evaluate the formulated framework. The experiments aim to measure the algorithmic efficiency (RQ1), map the economic trade-offs of the Pareto front (RQ2), and quantify strategic obfuscation introduced by the multi-objective ambiguity (RQ3).

A. Simulation Environment and Entities

Before systematically varying the battlefield conditions, a baseline simulation environment was established. The environment was populated with specific, heterogeneous object classes. This reflects modern Integrated Air Defense Systems (IADS), which layer expensive, high-reliability interceptors with combined with cheaper, low-tier point defenses for asset protection of varying strategic importance.

The specific parameters for the simulated weapon systems and protected ground assets are detailed in Table I and Table II, respectively. To accurately reflect the asymmetric saturation tactics, the incoming threat is modeled as a homogeneous swarm of Shahed-136 (SHA) loitering munitions [17] which are commonly used in modern conflicts around the world.

TABLE I
HETEROGENEOUS WEAPON SYSTEMS

System	Tactical Role	Cost C_i	Base p_{ij}	Cap. W_i
MIM-104 Patriot	High-Tier Area	\$100,000	0.85	4
9K35 Strela-10	Low-Tier Point	\$60,000	0.55	4

TABLE II
HETEROGENEOUS GROUND ASSETS

Asset Class	Description	Base Value V_k
Oil Refinery	High-Value Infrastructure	\$1,500,000
Headquarters (HQ)	Medium-Value Command Node	\$500,000

By introducing heterogeneity of assets and weapon systems, the NSGA-II solver is forced to navigate complex, non-linear economic decisions. It should force the solver to make tough decisions, e.g., whether an HQ justifies a Patriot interceptor, or if a cheaper Strela provides a more mathematically efficient marginal return, risking asset loss.

B. Procedural Generation and Experimental Design

To systematically analyze the framework, a One-Factor-At-A-Time (OFAAT) experimental design was employed [18]. A standard baseline scenario was established with a moderate swarm size ($|T| = 500$), a balanced resource deficit ($\rho = 0.50$), and uniform targeting of drones.

To ensure reproducibility, the procedural scenario generator followed strict structural rules. Defending batteries were spawned stochastically to meet the defined ρ capacity, utilizing a 40/60 probability split between Patriot and Strela systems, replicating IADS heterogeneity. Incoming threats were generated in autonomous flight clusters that adhere to real combat scenarios. During the evaluation of each Research Question, only the relevant independent variables described above were manipulated against the established baseline.

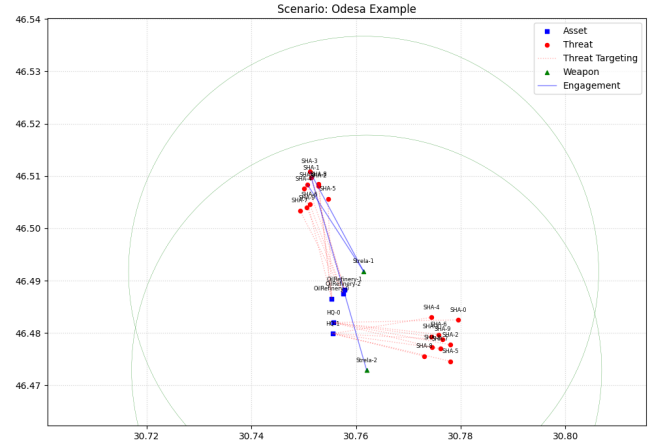


Fig. 1. Example of a procedurally generated AB-WTA engagement scenario within the framework, illustrating the distribution of ground assets, defending weapon batteries, and incoming loitering munition clusters.

1. Swarm Scale and Cluster Generation (RQ1): To test computational scalability, the total swarm size ($|T|$) has been varied across multiple tiers ($|T| \in 50, 100, 500, 1000$), built from autonomous clusters of 10 drones each. For these tests, resource scarcity and targeting density remained locked at their baseline values ($\rho = 0.50$, Uniform).

2. Resource Scarcity Ratios (RQ2): To force the “hold fire” mechanics to activate, the scenarios utilized a strict Ammo-to-Drone ratio ($\rho = \sum W_i / |T|$). The experiments evaluated severe deficits ($\rho = 0.25$), moderate deficits ($\rho = 0.50$), and near-parity scenarios ($\rho = 0.80$). Swarm size was fixed at $|T| = 500$ with a uniform swarm targeting strategy.

3. Targeting Density and Distribution (RQ3): To evaluate the model’s ability to obfuscate defenders’ intentions, both

the targeting distribution and asset heterogeneity were manipulated, while swarm size ($|T| = 100$) and scarcity ($\rho = 0.50$) were kept constant. The framework evaluated:

- *Uniform Targeting*: Incoming threats are distributed evenly across a completely homogeneous map of identical ground assets (i.e., all Headquarters), establishing a perfectly asset-wise homeogeneity baseline.
- *Concentrated Targeting*: In contrast, the map introduces strategic heterogeneity by mixing high-value Oil Refineries with standard Headquarters. The swarm deployment follows an asymmetric targeting distribution directing 80% of threats at the high-value hotspots, i.e., Oil Refineries asset clusters.

The resulting assignments are then analyzed to determine if the mathematical variance across the NSGA-II Pareto front successfully masks the true heterogeneous asset valuations under intense pressure.

C. Algorithm Configurations

NSGA-II Hyperparameters and Tiers: To thoroughly evaluate the trade-off between computational latency and Pareto front quality (RQ1), the NSGA-II solver was tested across three distinct hyperparameter configurations:

- *Fast Configuration*: Population Size = 20, Generations = 20.
- *Medium Configuration*: Population Size = 50, Generations = 50.
- *Heavy Configuration*: Population Size = 100, Generations = 200.

The crossover and mutation rates were abstracted to the `pymoo` library’s default adaptive configurations for NSGA-II.

Algorithmic Initialization: The initial population for the NSGA-II solver is generated purely stochastically. No seeding methods, such as injecting the Greedy MMR solution, were utilized in this framework. It is worth noting that the software implementation allows for seed injection, which was disabled during latency and result evaluation.

Baseline Configurations: The Greedy Maximum Marginal Return (MMR) heuristic requires no stochastic hyperparameter tuning, but is configured to iterate through a discrete range of 11 preference weights ($w \in [0.0, 0.1, \dots, 1.0]$), imitating a variety of objective valuation preferences, to generate a baseline Pareto front. To ensure parity, the Uniform Random Search baseline was configured to execute the exact same number of evaluations as the corresponding NSGA-II tier being benchmarked against it.

D. Experimental Procedure and Metrics

Because NSGA-II and the Uniform Random Search rely on stochastic processes, each scenario variation was independently simulated 5 times, which should result in less biased results and give more insight into actual patterns or tendencies.

To answer RQ1, execution times were recorded to benchmark the metaheuristic against the Greedy heuristic. For RQ2, the final non-dominated solution sets were extracted to visualize the Pareto front trade-offs between Engagement

Cost (F_1) and Expected Asset Loss (F_2), and their respective Hypervolumes were calculated to measure objective space coverage. Finally, to answer RQ3, specific assignments, mainly the purely kinetic Greedy solution ($w = 1.0$), alongside the median and a randomly sampled strategy from the NSGA-II Pareto front, were fed into the Informed Observer model. Spearman’s Rank Correlation Coefficient (r_s) between the true asset priorities and the adversary’s financial estimates was then calculated across all runs to measure strategic obfuscation of each approach.

IV. RESULTS

This section presents the empirical data generated by the AB-WTA framework across the simulated engagement scenarios. To ensure statistical reliability, all stochastic metrics represent the average of 5 independent simulation runs per parameter configuration.

A. Algorithmic Execution Latency (RQ1)

Table III details the average execution times required to process the defensive assignments across swarm scales ($|T|$).

TABLE III
AVERAGE EXECUTION TIME (SECONDS) BY SWARM SIZE ($\rho = 0.50$, UNIFORM TARGETING)

Solver Algorithm	$ T = 50$	$ T = 100$	$ T = 500$	$ T = 1000$
Greedy MMR	0.00s	0.00s	0.02s	0.06s
Random Search	9.21s	18.42s	94.17s	207.51s
NSGA-II (Fast)	0.32s	0.45s	1.91s	4.07s
NSGA-II (Medium)	1.48s	2.58s	11.73s	24.53s
NSGA-II (Heavy)	11.64s	20.52s	93.60s	195.05s

The deterministic Greedy heuristic recorded near-instantaneous execution times at all scenario scales. Among the metaheuristic configurations, the NSGA-II Fast configuration averaged 4.07 seconds at the maximum swarm tier ($|T| = 1000$), while the Heavy configuration recorded an average latency of 195.05 seconds under identical maximum load conditions. It can also be observed, that the Random Search performs slightly worse latency-wise to a corresponding NSGA-II configuration.

B. Objective Space and Hypervolume (RQ2)

To map the economic performance of the framework, the objective space plotting Engagement Cost (F_1) against Expected Asset Loss (F_2) was recorded. Figure 2 illustrates a representative sample of the non-dominated solution sets generated by the solvers during a severe deficit scenario ($\rho = 0.25$).

The generated Pareto curves visually span from the maximum ammunition allocation boundary down toward minimum financial expenditure. The NSGA-II metaheuristic produced a densely populated, continuous distribution of non-dominated compromise solutions. In contrast, the Greedy baseline produced a discrete set of coordinates rigidly bounded by its preference weights (w), having 3 distinct assignment options.

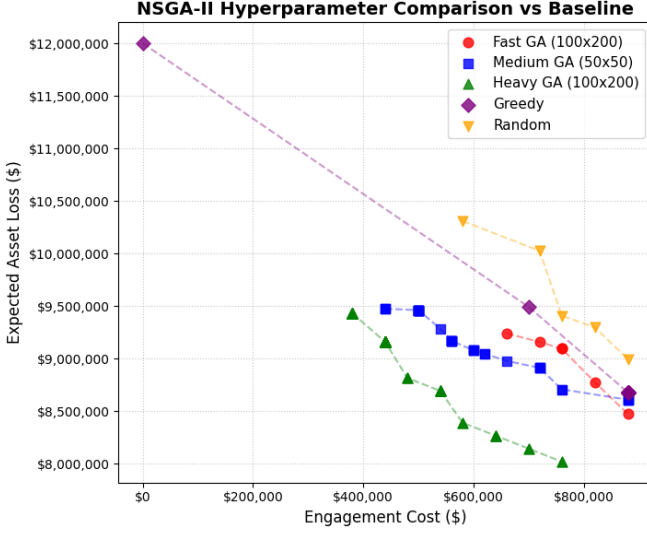


Fig. 2. Pareto fronts demonstrating the economic trade-off between Engagement Cost and Expected Asset Loss ($|T| = 50, \rho = 0.25$, Concentrated Targeting).

To quantify the objective space coverage of these trade-off menus, the Hypervolume (HV) indicator [19] was calculated against a theoretical worst-case reference point (representing maximum possible cost and total asset loss, augmented by a 1% buffer). Figure 3 presents the distribution of these HV scores across the evaluated resource scarcity tiers.

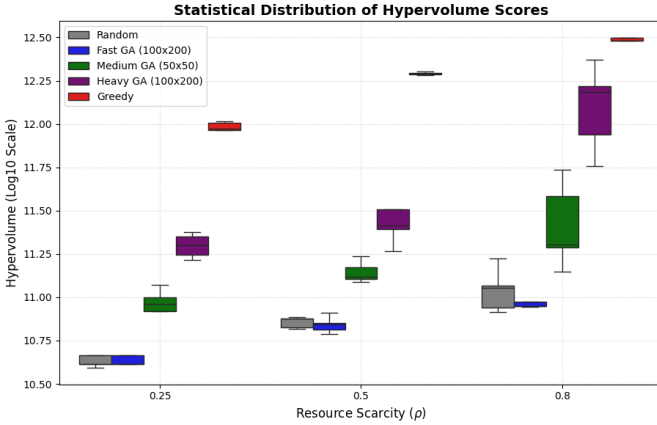


Fig. 3. Statistical distribution of Hypervolume scores across varying resource scarcity ratios ($\rho \in \{0.25, 0.50, 0.80\}$, $|T| = 500$, Uniform Targeting).

The empirical distributions reveal that the deterministic Greedy solver consistently yielded a higher median Hypervolume (HV) score than all counterparts across every evaluated scenario. Among the NSGA-II configurations, a clear mathematical hierarchy can be observed, with heavier configurations consistently generating greater objective space coverage than lighter configurations. Furthermore, it can be observed that as the resource scarcity ratio (ρ) increases, the overall hypervolume distribution correspondingly shifts upward across all

evaluated solvers.

C. Observer Model Variance and Strategic Ambiguity (RQ3)

Table IV reports the Spearman's Rank Correlation Coefficient (r_s) calculated using the Informed Observer model. This metric quantifies the statistical correlation between the true strategic hierarchy of the assets (V_k) and the adversary's financial estimates (E_k) based on observed interceptions.

TABLE IV
INFORMED OBSERVER SPEARMAN CORRELATION ($|T| = 100, \rho = 0.50$)

Assignment Strategy	Uniform (r_s)	Concentrated (r_s)
Greedy MMR ($w = 1.0$)	0.000	0.935
Random Baseline	0.000	0.548
NSGA-II Fast (Median Sample)	0.000	0.479
NSGA-II Fast (Random Sample)	0.000	0.512
NSGA-II Medium (Median Sample)	0.000	0.512
NSGA-II Medium (Random Sample)	0.000	0.505
NSGA-II Heavy (Median Sample)	0.000	0.332
NSGA-II Heavy (Random Sample)	0.000	0.396

Under the Uniform targeting distribution, all evaluated assignment strategies yielded a Spearman correlation of 0.000.

Under Concentrated targeting strategy, the correlation scores diverged across the evaluated algorithms. The deterministic Greedy MMR baseline produced the highest positive correlation with an average r_s of 0.935, while the Random baseline generated an average r_s of 0.548. Across the metaheuristic configurations, the NSGA-II Heavy solver recorded the lowest correlation metrics, yielding an average r_s of 0.332 when applying the median assignments and 0.396 when sampling randomly.

V. DISCUSSION

This section interprets results and evaluates the framework's practical viability.

A. Computational Scalability for Real-Time Operations (RQ1)

Execution latency remains a critical aspect in implementing effective WTA systems [8]. While the deterministic Greedy MMR maintained near-instantaneous execution speed under maximum load ($|T| = 1000$), the NSGA-II Fast configuration demonstrated highly competitive latency, resolving massive saturation scenarios well within actionable tactical timeframes.

The closely aligned execution times of Random Search and NSGA-II indicate that latency is driven primarily by raw objective function computations and/or population generation rather than NSGA-II's evolutionary overhead. Furthermore, the Random baseline marginally exceeded the Heavy GA's latency because the `pymoo` library is likely to skip execution of redundant fitness evaluations for unmodified surviving individuals, whereas Random Search evaluates every matrix from scratch.

Consequently, while the Heavy GA provides mathematically rigorous Pareto convergence, its latency scaling renders it impractical for real-time applications unless the aforementioned bottlenecks are resolved. For live AB-WTA utilization,

prioritizing a smaller, faster evolutionary population (Fast GA) is tactically superior to exhaustively mapping the objective space.

B. Economic Trade-offs and the Objective Space (RQ2)

Visual inspection of the Pareto fronts confirms the successful activation of the multi-objective mechanics, producing a wide variety of optimal assignments. However, the metaheuristics struggled to mathematically isolate the extreme \$0 engagement cost solution. This allowed the Greedy baseline to achieve superior Hypervolume (HV) simply by anchoring on these extreme edge cases.

Beyond the baseline comparison, a strict mathematical hierarchy is observed among the NSGA-II configurations, demonstrating that as the algorithm's evolutionary capacity expands, it consistently maps a larger area of the Pareto front. This indicates that the genetic operators do not plateau prematurely.

Furthermore, the positive correlation between the resource scarcity ratio (ρ) and the absolute HV scores highlights that as the available interceptor inventory expands, the combinatorial decision domain grows. This generates a significantly broader spectrum of valid, non-dominated defensive strategies for the solvers to discover, naturally driving the Hypervolume distribution upward across all evaluated configurations.

C. Strategic Ambiguity and Information Obfuscation (RQ3)

The empirical data from the Informed Observer model directly validates the hypothesis that deterministic defense models are vulnerable to predictable information leakage. Under concentrated saturation attacks, the Greedy heuristic rigidly prioritizes maximum kinetic return, resulting in a near-perfect Spearman correlation ($r_s = 0.936$). This deterministic predictability effectively allows a simulated adversary to accurately reverse-engineer the defender's hidden infrastructure hierarchy simply by observing assignments.

In contrast, the multi-objective framework successfully decoupled financial expenditure from actual asset valuation. By actively sampling diverse—yet optimal—solutions from the NSGA-II Pareto front, the model introduced significant strategic variance into the engagement patterns. This ambiguity severely degraded the adversary's predictive correlation ($r_s = 0.332$ for the median Heavy GA strategy). The data demonstrates that metaheuristic variance blinds the Informed Observer model, successfully obfuscating asset valuations without requiring the defender to abandon economic efficiency.

VI. CONCLUSION

This research formulated and evaluated a multi-objective Asset-Based Weapon-Target Assignment (AB-WTA) framework designed for air defense resource allocation optimization.

The empirical evaluation of the framework yielded three primary conclusions corresponding to the defined research objectives:

1. Computational Viability (RQ1): The results demonstrate that by utilizing a constraint-free, integer-based chromosome encoding, a minimal configuration of NSGA-II can

achieve highly competitive execution latencies compared to evaluated baselines. This indicates that multi-objective evolutionary algorithms can successfully operate within the strict real-time constraints required by modern Integrated Air Defense Systems (IADS). Ultimately, the study concludes that the true computational bottleneck lies in the stochastic population generation.

2. Economic Scaling under Scarcity (RQ2):

During severe ammunition deficit scenarios ($\rho = 0.25$), the genetic algorithm successfully mapped the non-linear economic trade-offs. By mathematically justifying "Hold Fire" decisions, the framework ensures that high-value assets remain protected even when the total interception of the incoming swarm is physically and logistically impossible.

3. Strategic Obfuscation (RQ3):

The data from the experiments highlights that the Heavy GA achieved lower predictability than the Random baseline. By activating "hold fire" mechanics during severe deficits, the NSGA-II solver deliberately sacrifices overwhelmingly saturated high-value assets, efficiently reallocating those interceptors to cheaper targets instead, while successfully obfuscating the asset valuation.

Societal Implications:

This research concludes that the proposed asset-aware optimization framework is well-suited for long-term attritional warfare. By dynamically balancing asset survival with ammunition conservation, the system prevents the depletion of interceptor stockpiles during asymmetric saturation attacks. Adopting these cost-aware assignment strategies is critical to ensuring the long-term economic sustainability of air defense operations in the modern battlefield.

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